

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD -402 103
Winter End Semester Examination – December 2019

Branch: Civil Engineering

Sem.:- III

Subject- Surveying-I

Subject Code (BTCVC304)

Marks: 60

Date:- 17/12/2019

Time:- 3 Hr.

Instructions to the Students:

1. Each question carries 12marks.
2. Solve **Any Five** questions of the following.
3. Illustrate your answers with neat sketches, diagram etc., wherever necessary.
4. If some part or parameter is noticed to be missing, you may appropriately assume it and should mention it clearly.

Q.1 Solve Any Two of the following.

12

- A) Two stations A and B are not intervisible due to rising ground between them. Explain with a neat sketch how the line AB can be ranged?
- B) What is mean by offset? List the instruments used for setting the offsets. Describe the construction of open cross staff with neat sketch.
- C) An old map was plotted to a scale of 40m to 1cm. Over the years, this map has been shrinking, and a line originally 200mm long is only 195mm long at present. Again the 20m chain was 5cm too long if the present area of the map measured by the planimeter is 125.50cm^2 , find the true area of the land surveyed.

Q.2 Solve Any Two of the following.

12

- A) The following bearings were observed where local attraction was suspected. Calculate the actual bearings.

Line	FB	BB
AB	$40^{\circ}30'SW$	$41^{\circ}15'NE$
BC	$80^{\circ}45'SW$	$79^{\circ}30'NE$
CD	$19^{\circ}30'NE$	$20^{\circ}00'SW$
DA	$80^{\circ}00'SE$	$80^{\circ}00'NW$

- B) Define the following terms:

- a) Whole circle bearing
- b) Local attraction
- c) Closing error

- C) In an anticlockwise traverse ABCA all sides were equal. Magnetic fore bearing of BC was obtained to be $15^{\circ}30'$. The bearing of sun was observed to be $184^{\circ}30'$ at local noon with a prismatic compass. Calculate the magnetic bearing and true bearing of all the sides of a traverse.

Q.3 Solve Any Two of the following.

12

- A) What is the principle of plane table surveying? List all accessories of Plane Table with a neat sketch and state use of each.
- B) Explain Methods of plan table
- i) Intersection Method
 - ii) Radiation Method
- C) How will you set plane table? Explain the procedure.

Q.4 Solve Any Two of the following.

12

- A) Define contour. Explain the characteristics of contour with neat sketch.
- B) During a fly levelling operation the following observations were made
- BS : 0.650, 2.155, 1.405, 2.655, 2.435m
- FS : 2.455, 1.305, 0.555, 2.405m
- The first back sight was taken on a BM of RL 90.500m. From the last back sight, it is required to set four pegs each at a distance of 30m on a falling gradient of 1 in 100. Calculate the RLs of these four pegs.
- C) A levelling instrument was set up exactly mid-way between two pegs A and B, 100m apart. The staff reading on A and B were 1.875 and 1.790 respectively. The instrument was then set up at a distance of 10m from A on the line AB. The respective staff readings were 1.630 and 1.560.
- Calculate the correct staff reading on A and B when the line of collimation was exactly horizontal.

Q.5 Solve Any Two of the following.

12

- A) i) What are the different checks in closed traverse in theodolite traversing.
- ii) How will you measure the horizontal angle by repetition method? Explain with an example.

B) The following observations were taken from stations P and Q. Calculate the length and bearing of AB, and also the angles PAB and QBA.

Line	Length(m)	Bearings
PA	125.0	60°30'SW
PQ	200.0	30°30'NE
QB	150.5	50°15'NW

C) The following records are obtained in a traverse survey, where the length and bearing of the last line were not recorded. Compute the length and bearing of line DA.

Line	Length (m)	Bearing
AB	75.50	30°24'
BC	180.50	110°36'
CD	60.25	210°30'
DA	?	?

Q.6 Solve the following.

12

A) Write short note on:

- Factors necessary for the selection of good alignment for a road.
- Setting out a building with an example.

B) Define curve, how it is designated? What are the different classification of a curve explain by sketch.

=====Paper End=====